Online Appendix to "Testing for Racial Bias in Police Traffic Searches"

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A Derivations

A.1 Deriving the random threshold in (1)

My model is an extension of the model by Canay et al. (2023) (henceforth, CMM). For convenience, I replicate the derivation of their model to better highlight the extension.

Let $Y_{si} \in \{0, 1\}$ denote the potential outcome of whether contraband is found on driver i when $Search_i = s$.[1] I assume that contraband is found if and only if the driver is carrying contraband and searched, which implies $Guilty_i = Y_{1i} - Y_{0i}$. Let $R_i \in \{w, m\}$ denote the race of the driver, V_i denote the characteristics of the driver observed by the officer and not the econometrician, and Z_i denote the instrument. Variables observed by both the officer and econometrician are implicitly conditioned on and I suppress their notation for brevity.

Let U_i denote a preference shock experienced by the officer when interacting with driver i. U_i is unobserved by the econometrician. The inclusion of U_i differentiates my model from that of CMM and, as shown below, permits a random threshold. Let $b(R_i, Z_i, V_i, U_i)$ denote

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[1]The methods extend to the case where Y_{si} is continuous.

the officer's taste for searching driver i.[2] Let C_i denote the cost of searching driver i, which represents other considerations besides bias. Let $Y^\star_{si} \in \{0,1\}$ denote the potential outcome the officer considers when making his decision, which can differ from the true Y_{si}. The officer solves the following problem of whether to search driver i,

$$\max_{s\in\{0,1\}} \mathcal{E}\left[Y^\star_{si} + s(b(R_i, Z_i, V_i, U_i) - C_i) \mid R_i = r, Z_i = z, V_i = v, U_i = u\right],$$

where $\mathcal{E}[\cdot \mid R_i = r, Z_i = z, V_i = v, U_i = u]$ denotes the officer's subjective conditional expectation. The solution to the officer's problem is

$$\mathit{Search}_i = \mathbb{1}\left\{\underbrace{\mathcal{E}[Y^\star_{1i} - Y^\star_{0i} \mid R_i = r, Z_i = z, V_i = v, U_i = u]}_{\text{Officer's perceived risk}} \geq -b(r,z,v,u) + c(r,z,v,u)\right\},$$

where

$$c(r,z,v,u) = \mathcal{E}[C_i \mid R_i = r, Z_i = z, V_i = v, U_i = u]$$

is the officer's perceived cost of searching the driver. Define

$$\begin{aligned}\lambda(r,z,v,u) \equiv\ & \underbrace{\mathbb{E}[Y_{1i} - Y_{0i} \mid R_i = r, Z_i = z, V_i = v, U_i = u]}_{\text{True risk, } \mathbb{P}\{\mathit{Guilty}_i \mid R_i=r, Z_i=z, V_i=v, U_i=u\}} \\ & - \underbrace{\mathcal{E}[Y^\star_{1i} - Y^\star_{0i} \mid R_i = r, Z_i = z, V_i = v, U_i = u]}_{\text{Officer's perceived risk}}\end{aligned}$$

to be the deviation between the true risk of the driver and the officer's risk assessment. The officer's search decision may then be written as

$$\mathit{Search}_i = \mathbb{1}\left\{\mathbb{P}\{\mathit{Guilty}_i \mid R_i = r, Z_i = z, V_i = v, U_i = u\} \geq \tau(r,z,v,u)\right\}, \tag{A.1}$$

[2] $b(\cdot)$ is denoted by $\beta(\cdot)$ in CMM.

where

$$\tau(r, z, v, u) \equiv -b(r, z, v, u) + c(r, z, v, u) + \lambda(r, z, v, u).$$

As in CMM, the threshold τ includes terms representing search tastes, search costs, and measurement error in risk. However, whereas the terms in the threshold in CMM depend only on R_i, Z_i, and V_i, I additionally allow the terms to depend on U_i. Consequently, the threshold is random even after conditioning on R_i, Z_i, and V_i. This allows the direction and intensity of bias to vary with the risk of the driver.

Assumption 1(ii) is satisfied by imposing the following assumption.

Assumption A1.

(i) $b(r, z, v, u) = b(r, u)$, $c(r, z, v, u) = c(r, u)$, *and* $\lambda(r, z, v, u) = \lambda(r, u)$.

(ii) $\mathbb{P}\{\textit{Guilty}_i \mid R_i = r, Z_i = z, V_i = v, U_i = u\} = \mathbb{P}\{\textit{Guilty}_i \mid R_i = r, Z_i = z, V_i = v\}$.

(iii) $U_i \perp\!\!\!\perp (Z_i, V_i) \mid R_i$.

Assumption A1(i) states the threshold is unaffected by the driver's non-race characteristics V_i, as in the Extended Roy Model of CMM, and is also unaffected by the instrument Z_i. The threshold T_i in the main paper may then be interpreted as

$$T_i = \tau(R_i, U_i) = b(R_i, U_i) + c(R_i, U_i) + \lambda(R_i, U_i).$$

Assumption A1(ii) states that the officer's preference shocks are not informative about risk after conditioning on driver characteristics and the IV, which implies $\tau(R_i, U_i) \perp\!\!\!\perp \textit{Guilty}_i \mid R_i, Z_i, V_i$.[3] Assumption A1(iii) states that the officer's preference shocks are independent of the driver's remaining characteristics after conditioning on the driver's race, which implies $\tau(R_i, U_i) \perp\!\!\!\perp (Z_i, V_i) \mid R_i$. Then by the contraction rule, $\tau(R_i, U_i) \perp\!\!\!\perp (\textit{Guilty}_i, Z_i, V_i) \mid R_i$, satisfying Assumption 1(ii).

[3]Under Assumption A1, $\lambda(r, u)$ depends on u solely through the officer's perceived risk.

A.2 Deriving the search and hit rates

The search rate is derived as follows.

$$\begin{aligned}
&\mathbb{E}[Search_i \mid R_i = r, Z_i = z] \\
&= \mathbb{E}[\mathbb{E}[Search_i \mid R_i = r, Z_i = z, V_i] \mid R_i = r, Z_i = z] \\
&= \mathbb{E}[\mathbb{E}[\mathbb{1}\{G(R_i, Z_i, V_i) \geq T_i\} \mid R_i = r, Z_i = z, V_i] \mid R_i = r, Z_i = z] \\
&= \mathbb{E}[F_{T|R}(G(r, z, V_i) \mid r) \mid R_i = r, Z_i = z] \\
&= \int_{\mathcal{V}} F_{T|R}(G(r, z, v) \mid r) \; dF_{V|R,Z}(v \mid r, z),
\end{aligned} \tag{A.2}$$

where the first equality is by law of iterated expectations; the second equality is by substituting the definition of $Search_i$; the third equality follows from $T_i \perp\!\!\!\perp (Z_i, V_i) \mid R_i$ imposed by Assumption 1; and the final equality follows by definition of conditional expectations.

The hit rate is derived as follows.

$$\begin{aligned}
&\mathbb{E}[Hit_i \mid R_i = r, Z_i = z] \\
&= \mathbb{E}[\mathbb{E}[Hit_i \mid R_i = r, Z_i = z, V_i] \mid R_i = r, Z_i = z] \\
&= \int_{\mathcal{V}} \mathbb{E}[Hit_i \mid R_i = r, Z_i = z, V_i = v] \; dF_{V|R,Z}(v \mid r, z),
\end{aligned} \tag{A.3}$$

where the first equality is by law of iterated expectations; and the second equality is by definition of conditional expectations. The expectation in the integrand may be written as

$$
\begin{aligned}
&\mathbb{E}[Hit_i \mid R_i = r, Z_i = z, V_i = v] \\
&= \mathbb{E}[Search_i \times Guilty_i \mid R_i = r, Z_i = z, V_i = v] \\
&= \mathbb{E}[Guilty_i \mid Search_i = 1, R_i = r, Z_i = z, V_i = v] \; \mathbb{E}[Search_i \mid R_i = r, Z_i = z, V_i = v] \\
&= \mathbb{E}[Guilty_i \mid G(r, z, v) > T_i, R_i = r, Z_i = z, V_i = v] \; \mathbb{E}[Search_i \mid R_i = r, Z_i = z, V_i = v] \\
&= \mathbb{E}[Guilty_i \mid R_i = r, Z_i = z, V_i = v] \; \mathbb{E}[Search_i \mid R_i = r, Z_i = z, V_i = v] \\
&= G(r, z, v) \; F_{T|R}(G(r, z, v) \mid r),
\end{aligned}
$$

where the first equality follows by definition of Hit_i; the second equality follows by law of iterated expectations, and that $Search_i \times Guilty_i = 0$ when $Search_i = 0$; the third equality follows from the definition of $Search_i$; the fourth equality follows from $T_i \perp\!\!\!\perp Guilty_i \mid R_i, Z_i, V_i$ from Assumption 1; and the final equality follows by definition of $G(\cdot, \cdot, \cdot)$, as well as from (A.2). Substituting this expression for $\mathbb{E}[Hit_i \mid R_i = r, Z_i = z, V_i = v]$ into (A.3) completes the derivation of the hit rate.

B Identifying and conducting inference on Θ

B.1 Identifying Θ

The bounds in Proposition 2 are sharp in the sense that they are the smallest and largest values of Θ. However, because bilinear programs are non-convex, Θ need not be the full interval derived in Proposition 2. Θ may be recovered by solving the following BP problem for $t \in [-1, 1]$,

$$\begin{aligned} Q^{\star}_{\theta}(t) \equiv & \min_{\boldsymbol{\omega}, \{\boldsymbol{\sigma}_r\}, \{\mathbf{p}_{r,z}\}} Q(\{\boldsymbol{\sigma}_r\}, \{\mathbf{p}_{r,z}\}) \\ \text{s.t.} \quad & \boldsymbol{\omega}'(\boldsymbol{\sigma}_m - \boldsymbol{\sigma}_w) = t, \ (9), (10), (11). \end{aligned}$$

The level of bias t is in Θ if and only if $Q^{\star}_{\theta}(t) = 0$.

B.2 Confidence intervals for Θ

The confidence interval for θ may be constructed by inverting the test for racial bias. To determine whether $t \in [-1, 1]$ is in the confidence interval, I first solve

$$\begin{aligned} \widehat{Q}^{\star}_{\theta}(t) \equiv & \min_{\boldsymbol{\omega}, \{\boldsymbol{\sigma}_r\}, \{\mathbf{p}_{r,z}\}} \widehat{Q}(\{\boldsymbol{\sigma}_r\}, \{\mathbf{p}_{r,z}\}) \\ \text{s.t.} \quad & \boldsymbol{\omega}'(\boldsymbol{\sigma}_m - \boldsymbol{\sigma}_w) = t, \ (9), (10), (11). \end{aligned}$$

I then construct the test statistic

$$\widehat{\tau}_{\theta}(t) = \widehat{Q}^{\star}_{\theta}(t) - \widehat{Q}^{\star}_{\text{Bias}}, \tag{B.4}$$

which compares the fit of the model when the officer is restricted to have bias $\theta = t$ against the fit when the officer is allowed to have any level of bias.

To estimate the distribution of $\widehat{\tau}_{\theta}(t)$ under the null hypothesis that $t \in \Theta$, I resample the data B times. For each resampled dataset, indexed by $b = 1, \dots, B$, I calculate (B.4) and denote it by $\widehat{\tau}_{\theta,b}(t)$. Then t does not enter the $(1-\alpha)$ confidence interval for θ if $\widehat{\tau}_{\theta}(t)$ exceeds the $1-\alpha$ quantile of $\{\widehat{\tau}_{\theta,b}(t) - \widehat{\tau}_{\theta}(t)\}$.

C Constraints in the bilinear programming problem

This section provides some examples of how to impose linear constraints in the bilinear program. This section also provides a numerical example motivating the monotonicity restriction (13) on the distributions of risk.

C.1 Imposing linear constraints

Consider the vector of variables $\mathbf{x} = (x_1, \ldots, x_K)'$. The monotonicity constraint

$$x_1 \leq x_2 \leq \cdots \leq x_K \tag{C.5}$$

may be written as

$$\begin{bmatrix} 1 & -1 & 0 & \ldots & 0 & 0 \\ 0 & 1 & -1 & \ldots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \ldots & 1 & -1 \end{bmatrix} \mathbf{x} \leq \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

To reverse the direction of monotonicity, simply reverse the inequalities. Linear constraints of the form

$$\sum_{k=1}^{K} a_k x_k \leq b$$

may be written as

$$\mathbf{a}'\mathbf{x} \leq b, \tag{C.6}$$

where $\mathbf{a} = (a_0, \ldots, a_K)'$.

To ensure that the search probabilities $\boldsymbol{\sigma}_r = (\sigma(g_0; r), \ldots, \sigma(g_K; r))$ that are being optimized over are consistent with being a cumulative distribution function of $T_i \mid R_i = r$ for $r \in \{w, m\}$, $\boldsymbol{\sigma}_r$ must be non-decreasing in index k, and each element must be in the unit interval. The non-decreasing property of $\boldsymbol{\sigma}_r$ takes the form of (C.5), and the bounds on each element of $\boldsymbol{\sigma}_r$ take the form of (C.6) (i.e., choose $\mathbf{a}$ to be a standard basis vector).

To ensure that each distribution of risk $\mathbf{p}_{r,z}$ is consistent with being a PMF, the elements of $\mathbf{p}_{r,z}$ must be in the unit interval and sum to 1. Both of these constraints take the form of (C.6). The researcher may also choose to impose monotonicity constraints on $\mathbf{p}_{r,z}$. These will take the form of (C.5).

If the researcher has a prior on how the average risk ranks across R_i and Z_i, the researcher can impose the ranking using linear constraints. To see how, write the average risk conditional on race and setting as

$$\begin{aligned}\mathbb{E}[\mathit{Guilty}_i \mid R_i = r, Z_i = z] &= \sum_{k=1}^{K} g_k \mathbf{p}_{r,z,k} \\ &= \mathbf{g}'\mathbf{p}_{r,z},\end{aligned}$$

where $\mathbf{g} = (g_0, \ldots, g_K)'$ is the vector of discretized risks. Then the ranking

$$\mathbb{E}[\mathit{Guilty}_i \mid R_i = r_1, Z_i = z_1] \leq \mathbb{E}[\mathit{Guilty}_i \mid R_i = r_2, Z_i = z_2]$$

takes the form

$$\begin{aligned} & \sum_{k=1}^{K} g_k \mathbf{p}_{r_1,z_1,k} \leq \sum_{k=1}^{K} g_k \mathbf{p}_{r_2,z_2,k} \\ \iff & \sum_{k=1}^{K} g_k \mathbf{p}_{r_1,z_1,k} - \sum_{k=1}^{K} g_k \mathbf{p}_{r_2,z_2,k} \leq 0 \\ \iff & \mathbf{g}'(\mathbf{p}_{r_1,z_1} - \mathbf{p}_{r_2,z_2}) \leq 0. \end{aligned}$$

This restriction has the same form as (C.6), with $\mathbf{a} = \mathbf{g}$ and $\mathbf{x} = \mathbf{p}_{r_1,z_1} - \mathbf{p}_{r_2,z_2}$.

C.2 Imposing integrality constraints

The BP framework nests earlier models in the literature where $Search_i = \mathbb{1}\{G_i \geq t(R_i)\}$ for some deterministic function t. These models effectively impose an integrality constraint on $\boldsymbol{\sigma}_r$ so that

$$\sigma(g_k; r) \in \{0, 1\} \text{ for } k = 1, \ldots K. \tag{C.7}$$

Under such a restriction, the BP program becomes a mixed integer program, which can also be solved to provable global optimality.

C.3 Motivating restrictions on the distribution of risk

In this section, I provide a simple numerical example for how the probability density function of risk for drivers stopped may be decreasing as risk increases, even though the officer may be more likely to stop drivers with higher risk.

Let $Stop_i \in \{0, 1\}$ denote the stop decision of an officer for driver i, and

$$\pi_{r,z}(g) \equiv \mathbb{P}\{Stop_i = 1 \mid R_i = r, Z_i = z, G_i = g\}$$

denote the probability that a driver is stopped conditional on her race R_i, the setting Z_i, and her risk G_i. Suppose for some $(r, z) \in \{w, m\} \times \mathcal{Z}$, that $\pi_{r,z}(g)$ is as shown in the top panel of Figure C.1. The officer has a 1% probability of stopping a driver with zero risk, and this probability monotonically increase to 50% as risk increases to unity.

Let $f^{Pop}_{G|R,Z}(\cdot \mid r, z)$ denote the density of risk among the population of drivers, i.e., the distribution of risk unconditional on being stopped. Suppose that the population risk follows a beta distribution with shape parameters 1 and 9, as depicted in the middle panel of Figure C.1. This implies 10% of all drivers carry contraband.

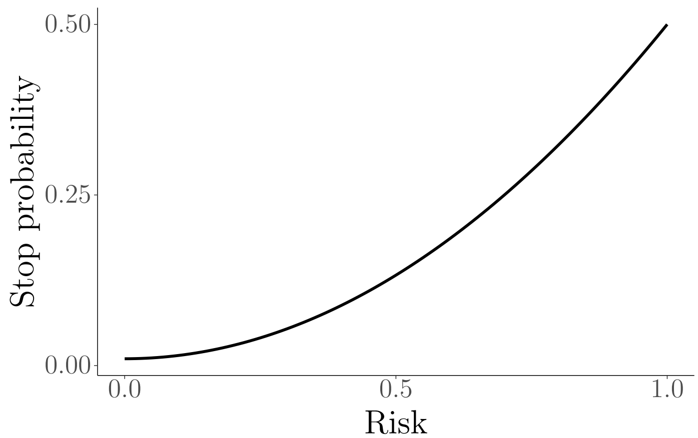
Then by Bayes' rule, the distribution of risk conditional on being stopped is

$$f_{G|R,Z}(g \mid r, z) = \frac{\pi_{r,z}(g)\ f^{Pop}_{G|R,Z}(g \mid r, z)}{\int_0^1 \pi_{r,z}(g')\ f^{Pop}_{G|R,Z}(g' \mid r, z)\ dg'},$$

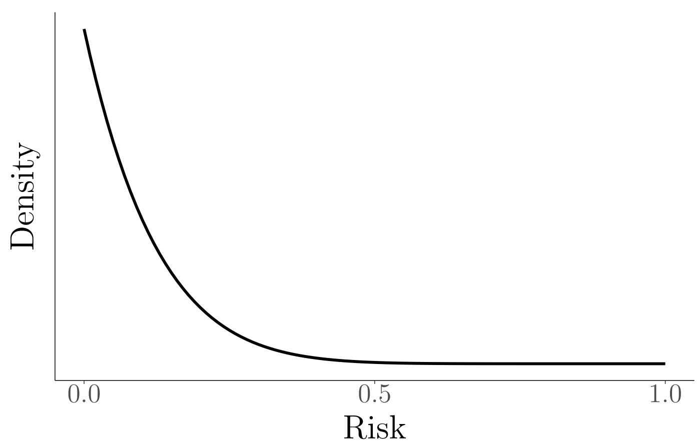
and is shown in the bottom panel of Figure C.1. Despite the officer being much more likely to stop high-risk drivers, the proportion of low-risk drivers in population is sufficiently large such that the density of risk conditional on being stopped is strictly decreasing.

Figure C.1: Monotone-decreasing density for risk of drivers stopped

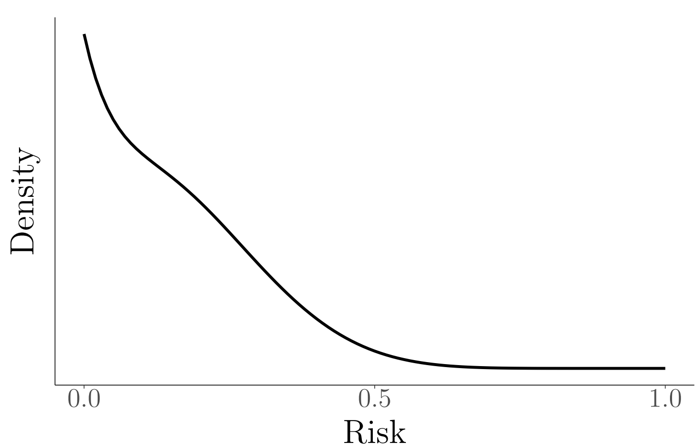
(a) Probability of stopping a driver



(b) Population distribution of risk



(c) Sample distribution of risk



D Modeling continuous risk using B-splines

In this section, I show how to adapt the methodology to allow for continuous distributions of risk.

Recall that the search and (unconditional) hit rates may be written as

$$\mathbb{P}\{Search_i = 1 \mid R_i = r, Z_i = z\} = \int_0^1 \sigma(g; r)\ dF_{G|R,Z}(g \mid r, z), \tag{D.8}$$

$$\mathbb{P}\{Hit_i = 1 \mid R_i = r, Z_i = z\} = \int_0^1 g\ \sigma(g; r)\ dF_{G|R,Z}(g \mid r, z). \tag{D.9}$$

Suppose there exists a density function $f_{G|R,Z}(\cdot \mid r, z)$ for $(r, z) \in \{w, m\} \times \mathcal{Z}$. Suppose also that $\sigma(g; r)$ and $f_{G|R,Z}$ can be modeled using B-splines. Then (D.8)–(D.9) can be written as bilinear terms

$$\mathbb{P}\{Search_i = 1 \mid R_i = r, Z_i = z\} = \boldsymbol{\sigma}_r' \mathbf{Q}_S \mathbf{p}_{r,z}, \tag{D.10}$$

$$\mathbb{P}\{Hit_i = 1 \mid R_i = r, Z_i = z\} = \boldsymbol{\sigma}_r' \mathbf{Q}_H \mathbf{p}_{r,z}, \tag{D.11}$$

where $\{\boldsymbol{\sigma}_r\}$, $\{\mathbf{p}_{r,z}\}$ are sets of parameters characterizing the officer's threshold distributions and the distributions of risk, respectively; and $\mathbf{Q}_S$ and $\mathbf{Q}_H$ are known matrices. Equations (D.10)–(D.11) follow from the fact that products of B-splines are also B-splines. Mørken (1991) provides the formula for calculating the coefficients for the products of two B-splines. To state this formula, it is necessary to first define several terms.

Following the notation in Mørken (1991), let k be a positive integer denoting the order of a spline, and $\boldsymbol{\tau} = (\tau_1, \tau_2, \dots)$ be a non-decreasing sequence of real numbers denoting the knots of the spline. Then the B-spline $B_{i,k,\boldsymbol{\tau}}$ is defined using the recurrence relation

$$B_{i,k,\boldsymbol{\tau}}(x) \equiv \omega_{i,k,\boldsymbol{\tau}}(x) B_{i,k-1,\boldsymbol{\tau}}(x) + (1 - \omega_{i+1,k,\boldsymbol{\tau}}(x)) B_{i+1,k-1}(x),$$

where

$$\omega_{i,k,\boldsymbol{\tau}}(x) \equiv \begin{cases} \frac{(x-\tau_i)}{\tau_{i+k-1}-\tau_i} & \text{if } \tau_i < \tau_{i+k-1}, \\ 0 & \text{otherwise,} \end{cases} \tag{D.12}$$

and

$$B_{i,1,\boldsymbol{\tau}}(x) \equiv \begin{cases} 1 & \text{if } \tau_i \le x < \tau_{i+1} \\ 0 & \text{otherwise.} \end{cases}$$

Let $\mathcal{S}_{k,\boldsymbol{\tau}}$ denote the linear space spanned by the splines $\{B_{i,k,\boldsymbol{\tau}}\}$.

Let $\boldsymbol{\tau}'$ be a subsequence of $\boldsymbol{\tau}$. Then $\mathcal{S}_{k,\boldsymbol{\tau}'} \subseteq \mathcal{S}_{k,\boldsymbol{\tau}}$, and the B-splines $\{B_{j,k,\boldsymbol{\tau}'}\}$ are linear combinations of the B-splines $\{B_{i,k,\boldsymbol{\tau}}\}$ and can be written as

$$B_{j,k,\boldsymbol{\tau}'} = \sum_i \alpha_{j,k,\boldsymbol{\tau}',\boldsymbol{\tau}}(i) B_{i,k,\boldsymbol{\tau}}.$$

The coefficients $\{\alpha_{j,k,\boldsymbol{\tau}',\boldsymbol{\tau}}\}$ are discrete B-spline of order k on $\boldsymbol{\tau}$ with knots $\boldsymbol{\tau}'$ and satisfy the recurrence relation

$$\alpha_{j,k,\boldsymbol{\tau}',\boldsymbol{\tau}}(i) = \omega_{j,k,\boldsymbol{\tau}'}(\tau_{i+k-1})\alpha_{j,k-1}(i) + (1 - \omega_{j+1,k,\boldsymbol{\tau}'}(\tau_{i+k-1}))\alpha_{j+1,k-1}(i),$$

where $\omega_{j,k,\boldsymbol{\tau}'}$ is defined as in (D.12), and $\alpha_{j,1,\boldsymbol{\tau}',\boldsymbol{\tau}}(i) = B_{j,1,\boldsymbol{\tau}'}(\tau_i)$.

Suppose we have two splines, $f_1 \in \mathcal{S}_{k_1,\boldsymbol{\tau}_1}$ and $f_2 \in \mathcal{S}_{k_2,\boldsymbol{\tau}_2}$. Let $\mathcal{S}_{k,\boldsymbol{\tau}}$ denote the spline space containing the product of f_1 and f_2. The order of this spline space is $k = k_1 + k_2 - 1$. The knots of this spline space $\boldsymbol{\tau}$ contains all the distinct knots in $\boldsymbol{\tau}_1$ and $\boldsymbol{\tau}_2$, with the multiplicity of the knots being determined as follows. For each knot τ in $\boldsymbol{\tau}$, let m_1 denote the multiplicity of τ in $\boldsymbol{\tau}_1$ and m_2 denote the multiplicity of τ in $\boldsymbol{\tau}_2$. Then the multiplicity of τ in $\boldsymbol{\tau}$ is

$$\widehat{m} = \begin{cases} \max(k_1 - 1 + m_2, k_2 - 1 + m_1) & \text{if } m_1 > 0 \text{ and } m_2 > 0, \\ k_1 - 1 + m_2 & \text{if } m_1 = 0 \text{ and } m_2 > 0, \\ k_2 - 1 + m_1 & \text{if } m_1 > 0 \text{ and } m_2 = 0, \\ 0 & \text{if } m_1 = 0 \text{ and } m_2 = 0. \end{cases} \tag{D.13}$$

Finally, let $P = \{p_1, \ldots p_{k_1-1}\}$ be a set of $k_1 - 1$ integers from $I_{k-1} = \{1, \ldots, k-1\}$. Let $Q = I_{k-1} \setminus P = (q_1, \ldots, q_{k_2-1})$ be the set of the remaining $k_2 - 1$ integers. For a given integer i, define the knot vectors $\boldsymbol{\tau}^P$ and $\boldsymbol{\tau}^Q$ by

$$\boldsymbol{\tau}^P = (\ldots, \tau_{i-1}, \tau_i, \tau_{i+p_1}, \tau_{i+p_2}, \ldots, \tau_{i+p_{k_1-1}}, \tau_{i+k}, \tau_{i+k+1}, \ldots), \tag{D.14}$$

$$\boldsymbol{\tau}^Q = (\ldots, \tau_{i-1}, \tau_i, \tau_{i+q_1}, \tau_{i+q_2}, \ldots, \tau_{i+q_{k_2-1}}, \tau_{i+k}, \tau_{i+k+1}, \ldots). \tag{D.15}$$

Let Π denote the set of all subsets of I_{k-1} consisting of $k_1 - 1$ elements.

Theorem 1. *(Theorem 3.1 of Mørken (1991)) Let $f_1 = \sum_{j_1} c_{1,j_1} B_{j_1,k_1,\boldsymbol{\tau}_1}$ and $f_2 = \sum_{j_2} c_{2,j_2} B_{j_2,k_2,\boldsymbol{\tau}_2}$ be two given spline functions. Set $k = k_1 + k_2 - 1$ and construct the knot vector $\boldsymbol{\tau}$ according to (D.13). Then $f_1 f_2 \in \mathcal{S}_{k,\boldsymbol{\tau}}$ so that there exists coefficients $d_1, d_2, \ldots$ such that $f_1(x) f_2(x) = \sum_i d_i B_{i,k,\tau}(x)$. Specifically, for a given i, the knot vectors $\boldsymbol{\tau}^P$ and $\boldsymbol{\tau}^Q$ defined by (D.14)–(D.15) satisfy $\boldsymbol{\tau}_1 \subseteq \boldsymbol{\tau}^P$ and $\boldsymbol{\tau}_2 \subseteq \boldsymbol{\tau}^Q$, and d_i is given by*

$$d_i = \sum_{P \in \Pi} \sum_{j_1} \sum_{j_2} c_{1,j_1} \alpha_{j_1,k_1,\boldsymbol{\tau}_1,\boldsymbol{\tau}^P}(i) c_{2,j_2} \alpha_{j_2,k_2,\boldsymbol{\tau}_2,\boldsymbol{\tau}^Q}(i) \Big/ \binom{k-1}{k_1-1}.$$

It follows from Theorem 1 that the integral of $f_1 f_2$ can be written as a bilinear term,

$$\int f_1(x) f_2(x)\, dx = \sum_{j_1} \sum_{j_2} c_{1,j_1} c_{2,j_2} v_{j_1,j_2},$$

where

$$v_{j_1,j_2} = \sum_i \sum_{P\in\Pi} \alpha_{j_1,k_1,\boldsymbol{\tau}_1,\boldsymbol{\tau}^P}(i)\alpha_{j_2,k_2,\boldsymbol{\tau}_2,\boldsymbol{\tau}^Q}(i) \int B_{i,k,\boldsymbol{\tau}}(x)\; dx \Bigg/ \binom{k-1}{k_1-1}$$

can be calculated. Equation (D.10) follows from letting f_1 be the threshold distribution $\sigma(\cdot;r)$, and letting f_2 be the density of risk $f_{G|R,Z}(\cdot \mid r,z)$. Equation (D.11) follows from the same reasoning, except f_1 is the threshold distribution scaled by risk, $g\sigma(g;r)$.[4]

Shape restrictions on σ and $f_{G|R,Z}(\cdot \mid r,z)$ may be imposed through an auditing procedure (Shea and Torgovitsky, 2023). This procedure consists of first imposing the shape constraints on a coarse *constraint* grid over the domains of $\sigma(\cdot;r)$ and $f_{G|R,Z}(\cdot \mid r,z)$. The BP problems is then solved. Whether the solutions for $\sigma(\cdot;r)$ and $f_{G|R,Z}(\cdot \mid r,z)$ satisfy the shape constaints is then checked on a much finer *audit* grid. Points in the audit grid where the shape constraints are violated are added to the constraint grid, and the BP problem is solved again. This process is repeated until the shape constraints are satisfied on all points of the audit grid. This procedure avoids the computational and mathematical difficulties of determining whether the B-splines satisfy properties such as monotonicity, boundedness, and convexity (De Boor, 2001).

[4] If $\sigma(g;r)$ is a B-spline, then $g\sigma(g;r)$ will also be a B-spline.

E Applying the test of Knowles et al. (2001)

For comparison, I apply the test of Knowles et al. (2001) to the 50 officers evaluated in the main paper. Table 1 presents the hit rates of white and minority drivers who are searched, as well as the p-values from Pearson χ^2 tests of equal hit rates. Under the model of Knowles et al. (2001), the null hypothesis of equal hit rates corresponds to officers being unbiased. Bold entries indicate the group of drivers that officers (as a collective) are biased against whenever bias is detected. Interestingly, officers appear to be biased against white drivers in most cases where bias is detected. This difference stems from how the test of Knowles et al. (2001) compares the police department-wide hit rate across white and minority drivers, whereas the method I propose compares both the search and hit rates in multiple settings across white and minority drivers for each officer. Because the test of Knowles et al. (2001) is derived from an equilibrium model, it is not intended to evaluate each officer separately and cannot identify which officers are biased. Nevertheless, the test detects bias in certain precincts and days/shifts, which potentially narrows down which officers are biased (under an equilibrium model).[5]

[5]It is still difficult to determine which officer is biased as many officers make traffic stops at each combination of precinct and day/shift.

Table 1: Knowles et al. (2001) test results

	Male and female drivers			Male drivers			Female drivers		
	White	Minority	p-value	White	Minority	p-value	White	Minority	p-value
All	**0.145**	0.174	0.000***	**0.151**	0.179	0.000***	**0.134**	0.158	0.005***
Precinct 1	0.186	0.192	0.779	0.187	0.172	0.505	**0.180**	0.287	0.039**
Precinct 2	0.206	0.217	0.495	0.202	0.215	0.5	0.214	0.224	0.74
Precinct 3	0.182	**0.144**	0.008***	0.171	0.150	0.204	0.210	**0.112**	0.001***
Precinct 4	**0.022**	0.083	0.000***	**0.022**	0.097	0.000***	**0.022**	0.051	0.068*
Precinct 5	0.203	**0.183**	0.054*	0.208	**0.183**	0.046**	0.193	0.183	0.618
Precinct 6	**0.167**	0.238	0.015**	**0.160**	0.240	0.033**	0.177	0.234	0.249
Precinct 7	0.183	**0.161**	0.073*	0.186	0.175	0.503	0.179	**0.127**	0.011**
Precinct 8	**0.053**	0.125	0.000***	**0.064**	0.138	0.000***	**0.035**	0.093	0.000***
Weekday morning	0.262	**0.227**	0.024**	0.267	**0.232**	0.057*	0.251	0.206	0.135
Weekday evening	0.177	0.183	0.55	0.184	0.185	0.928	0.164	0.174	0.528
Weekday night	**0.108**	0.165	0.000***	**0.114**	0.170	0.000***	**0.096**	0.146	0.000***
Weekend morning	0.266	**0.127**	0.014**	0.163	0.152	0.873	0.476	**0.023**	0.000***
Weekend evening	**0.104**	0.160	0.002***	**0.110**	0.162	0.014**	**0.091**	0.150	0.063*
Weekend night	**0.056**	0.106	0.000***	**0.055**	0.109	0.000***	**0.059**	0.100	0.017**

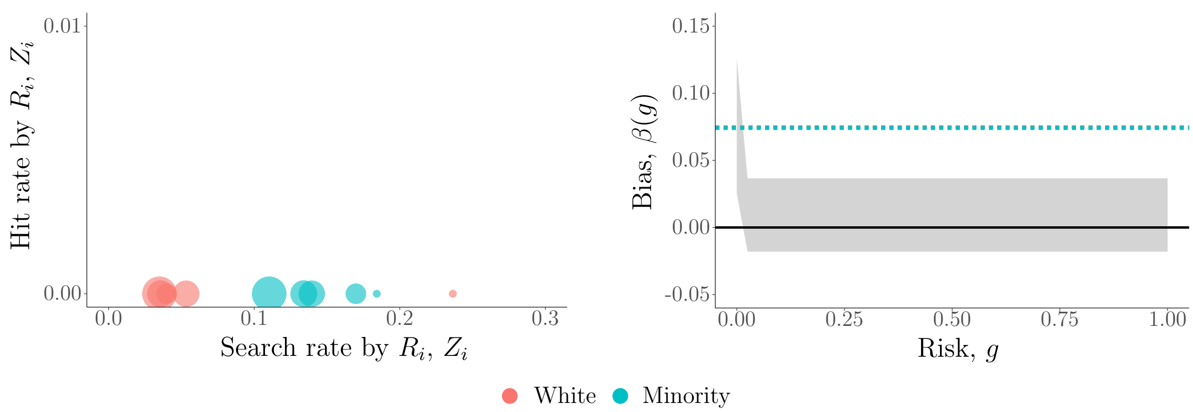
Notes: In each vertical panel, the first two columns present the hit rates for white and minority drivers who are searched. The third column presents the p-value from a Pearson χ^2 test for equal hit rates, conditional on certain characteristics of the driver and traffic stop. Bold hit rates indicate the group of drivers which officers are biased against. * 10%, ** 5%, *** 1% significance.

F Estimates for biased officers

F.1 Estimates when averaging search and hit rates over $X_i \mid R_i = w$

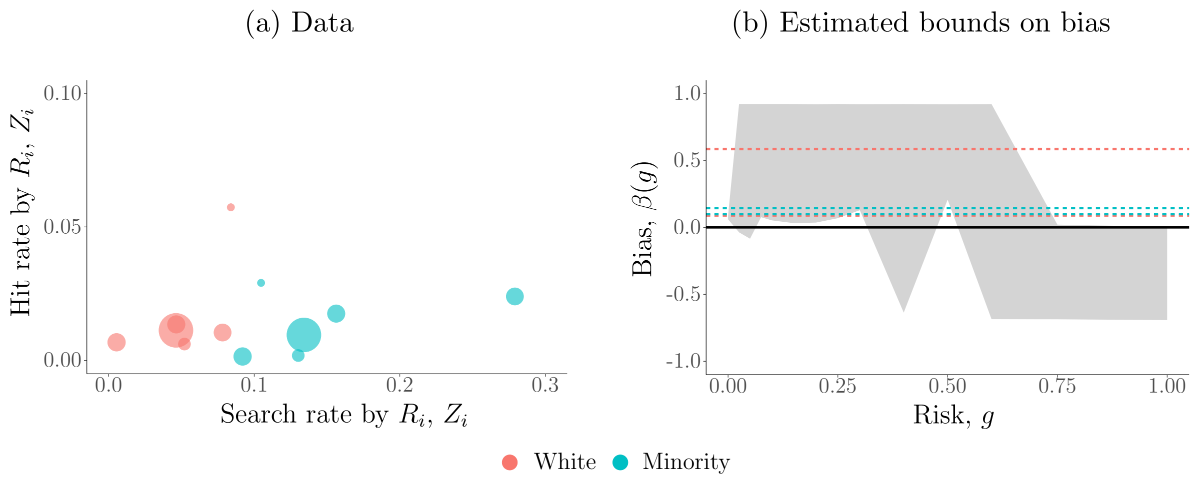
Figure F.2: Officer 8

(a) Data　　(b) Estimated bounds on bias



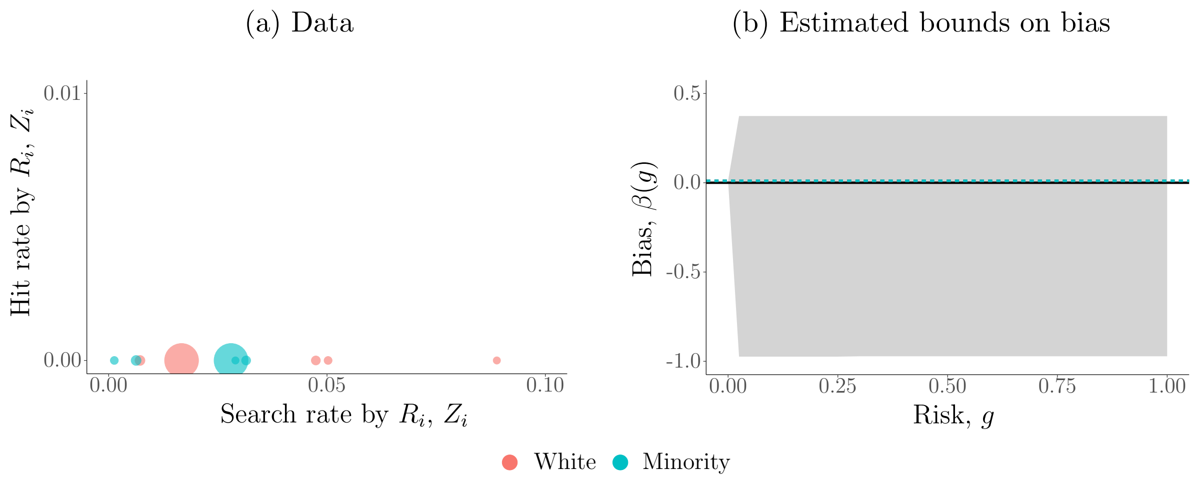
Note: The size of the dots in the left panel represents the number of stops at each setting. The dashed lines in the right panel indicate the bounds on the average bias. If white drivers were treated as minority drivers, holding their risk constant, they would be searched between 7.50 and 7.53 percentage points more on average. If minority drivers were treated as white drivers, holding their risk constant, they would be searched between 7.36 and 7.46 percentage points less on average.

Figure F.3: Officer 23



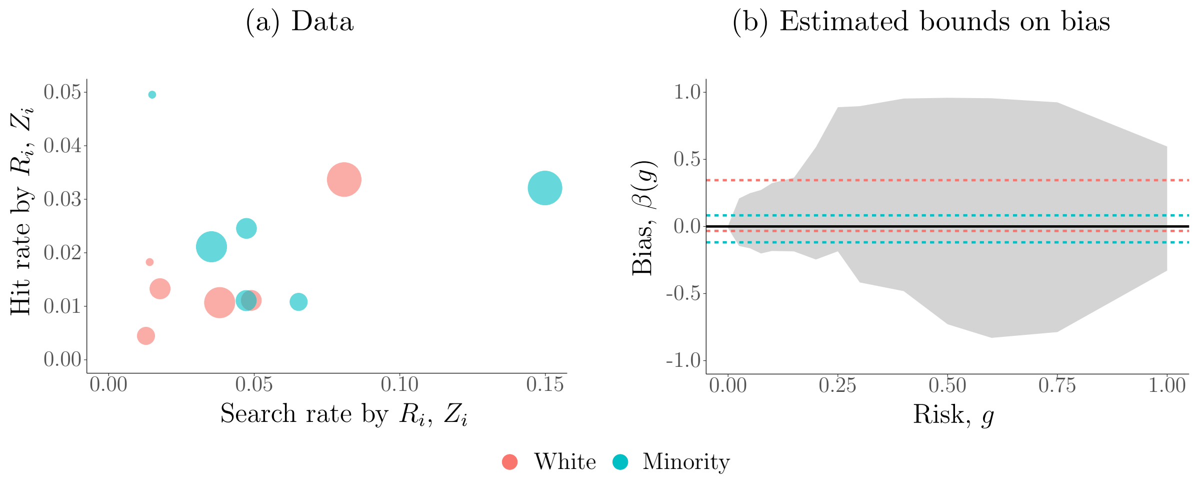
Note: The size of the dots in the left panel represents the number of stops at each setting. The dashed lines in the right panel indicate the bounds on the average bias. If white drivers were treated as minority drivers, holding their risk constant, they would be searched between 8.85 and 58.54 percentage points more on average. If minority drivers were treated as white drivers, holding their risk constant, they would be searched between 9.87 and 14.39 percentage points less on average.

Figure F.4: Officer 35



Note: The size of the dots in the left panel represents the number of stops at each setting. The dashed lines in the right panel indicate the bounds on the average bias. If white drivers were treated as minority drivers, holding their risk constant, they would be searched between 1.11 and 1.14 percentage points more on average. If minority drivers were treated as white drivers, holding their risk constant, they would be searched between 1.07 and 1.13 percentage points less on average.

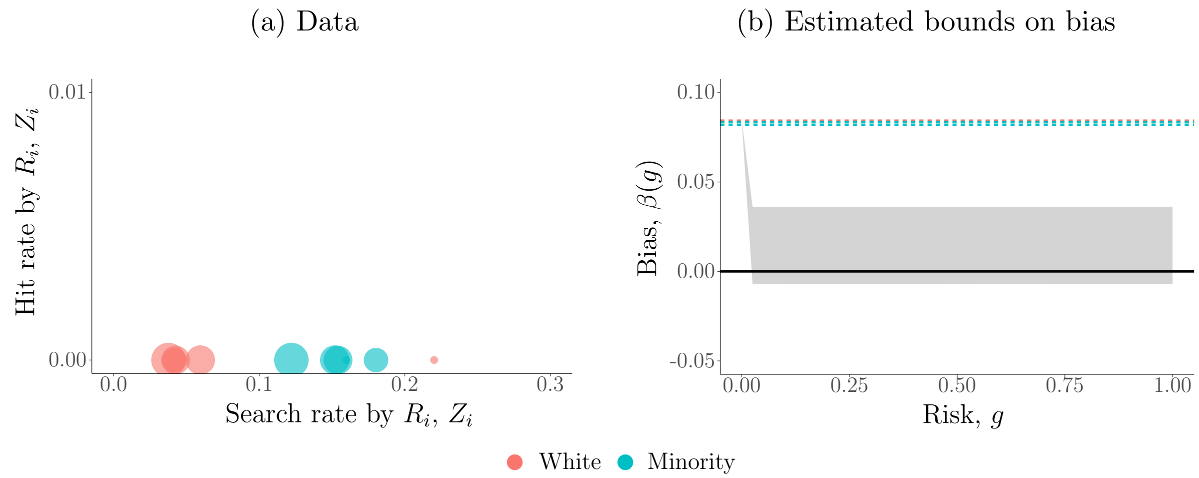
Figure F.5: Officer 41



Note: The size of the dots in the left panel represents the number of stops at each setting. The dashed lines in the right panel indicate the bounds on the average bias. If white drivers were treated as minority drivers, holding their risk constant, they would be searched between 3.34 percentage points less and 34.49 percentage points more on average. If minority drivers were treated as white drivers, holding their risk constant, they would be searched between 11.82 percentage points less and 8.28 percentage points more on average.

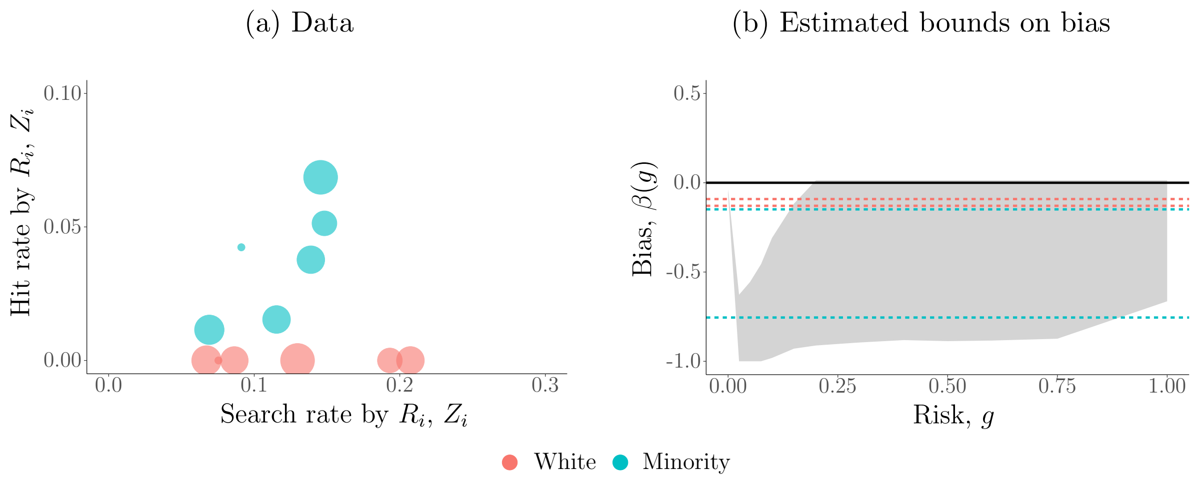
F.2 Estimates when averaging search and hit rates over $X_i \mid R_i = m$

Figure F.6: Officer 8



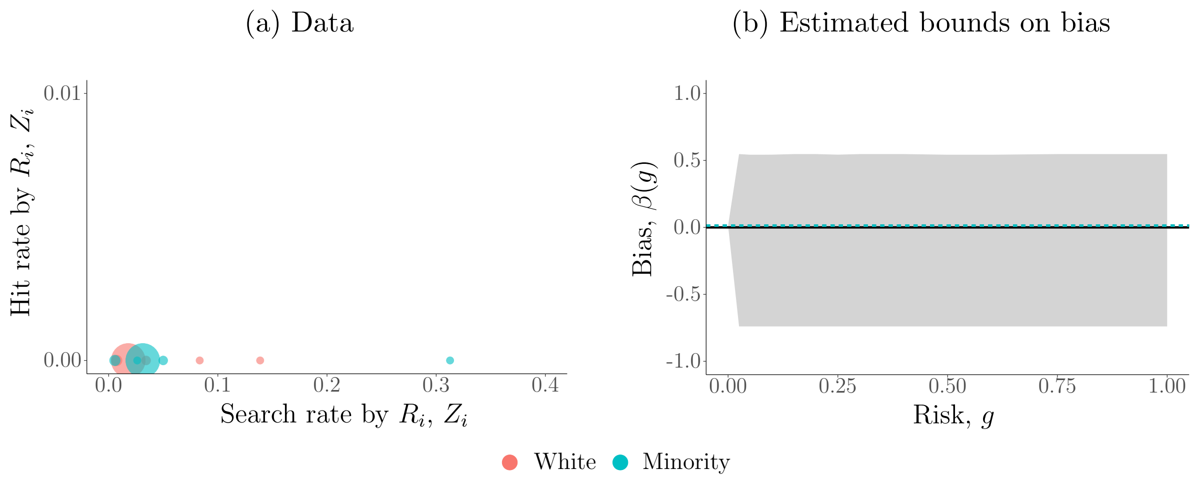
Note: The size of the dots in the left panel represents the number of stops at each setting. The dashed lines in the right panel indicate the bounds on the average bias. If white drivers were treated as minority drivers, holding their risk constant, they would be searched between 8.37 and 8.42 percentage points more on average. If minority drivers were treated as white drivers, holding their risk constant, they would be searched between 8.21 and 8.35 percentage points less on average.

Figure F.7: Officer 20



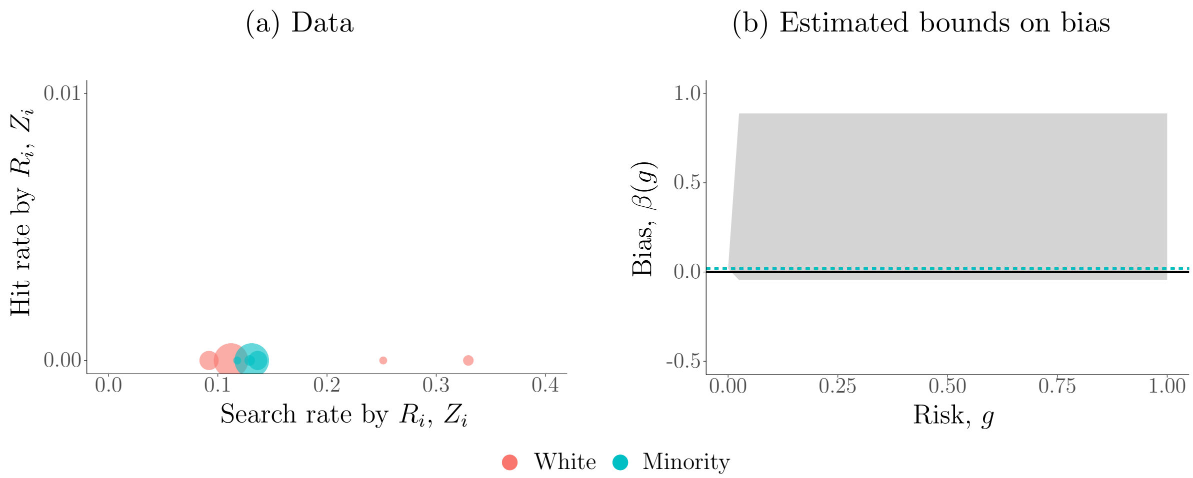
Note: The size of the dots in the left panel represents the number of stops at each setting. The dashed lines in the right panel indicate the bounds on the average bias. If white drivers were treated as minority drivers, holding their risk constant, they would be searched between 9.22 and 12.91 percentage points less on average. If minority drivers were treated as white drivers, holding their risk constant, they would be searched between 17.26 and 74.75 percentage points more on average.

Figure F.8: Officer 35



Note: The size of the dots in the left panel represents the number of stops at each setting. The dashed lines in the right panel indicate the bounds on the average bias. If white drivers were treated as minority drivers, holding their risk constant, they would be searched between 1.33 and 1.34 percentage points more on average. If minority drivers were treated as white drivers, holding their risk constant, they would be searched between 1.31 and 1.34 percentage points less on average.

Figure F.9: Officer 43



Note: The size of the dots in the left panel represents the number of stops at each setting. The dashed lines in the right panel indicate the bounds on the average bias. If white drivers were treated as minority drivers, holding their risk constant, they would be searched between 1.90 and 1.91 percentage points more on average If minority drivers were treated as white drivers, holding their risk constant, they would be searched between 1.90 and 1.99 percentage points less on average.

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